Distribution-Preserving QIM: Trading Noise Margin for Statistical Undetectability and Scaling Robustness in Invisible Watermarking

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***Abstract*—Quantization index modulation (QIM) watermarks write DCT coefficients onto a lattice, leaving a discrete histogram fingerprint of the kind that black-box removal attacks such as UnMarker search for. We study *Distribution-Preserving QIM* (DP-QIM), which partitions the natural distribution of a DCT coefficient into equal-mass cells and, when a coefficient must change cells to carry its bit, replaces it with a natural sample from the target cell. We compare DP-QIM with classical QIM at the same DCT position and matched PSNR on 44 crops from 15 real photographs, using source-grouped cross-validation and bootstrap confidence intervals. DP-QIM is close to undetectable: a learned histogram detector reaches ROC-AUC $0.52\,[0.49, 0.57]$ against DP-QIM versus $0.90\,[0.78, 1.00]$ against QIM at 64 bits, and 0.61 versus 1.00 at 256 bits, while the UnMarker peak statistic stays at chance. Because one-bit DP-QIM cells split at the coefficient median, which is exactly zero for the AC coefficient we use, decoding is a sign test that is invariant to amplitude scaling, giving $+35$ pp bit accuracy under Gaussian blur, $+34$ pp under $0.5\times$ downscaling and $+20$ pp under $0.5\times$ contrast scaling. The price is noise margin: QIM is more robust to JPEG (Q=90: -15 pp; Q=70: -28 pp) and additive noise (-14 pp), and an UnMarker-style removal attack degrades DP-QIM 3 to 7 pp more than QIM. Statistical undetectability alone therefore does not confer removal resistance. Code and the full evaluation protocol are released.**[1]

***Index Terms*—invisible watermarking, quantization index modulation, DCT, steganalysis, valumetric scaling, AI-generated content provenance**

I. Introduction

Invisible watermarks are a primary tool for tracing the provenance of images, including AI-generated ones [1]. Recent work shows they are fragile against adaptive adversaries. UnMarker [2] removes deployed watermarks without access to the detector or the key by locating spectral regions whose statistics differ from those of natural images and perturbing them, and regeneration attacks remove watermarks by re-synthesizing the image [3], [4].

Quantization index modulation (QIM) [5] and its descendants are especially exposed to the first kind of attack. QIM writes each carrier coefficient onto one of two interleaved lattices, so the marginal histogram of watermarked coefficients collapses onto isolated spikes (Fig. 1, middle), a fingerprint natural images do not have. This paper asks a narrow question: *if a QIM-style watermark is constrained to leave the coefficient marginal natural, what does it gain and what does it lose?*

We study Distribution-Preserving QIM (DP-QIM). It partitions the empirical distribution of a coefficient into $2M$ equal-mass cells with alternating bit labels; a coefficient already in a cell carrying its bit is left unchanged, and otherwise it is replaced by a natural sample drawn from the interior of an adjacent cell. Our contributions are:

1) The DP-QIM embedding rule and two properties: the watermarked marginal is a two-component mixture of natural distributions (Prop. 1), and one-bit DP-QIM decoding is invariant to amplitude scaling, whereas lattice decoding is not (Prop. 2).
2) A leakage-free comparison protocol for quantization watermarks: both methods at the same DCT position, robustness curves compared at matched PSNR, cross-validation grouped by source photograph, an attacker whose statistics are independent of the embedder's, and bootstrap confidence intervals (Sec. IV).
3) An empirical characterization of the trade-off on real photographs: near-chance detectability and large gains under scaling-type attacks, paid for with reduced margin against requantization and additive noise (Sec. V).

II. Related Work

QIM and its variants. QIM [5] and dither modulation are provably good under additive noise [6] but fragile to valumetric (amplitude) scaling. Rational dither modulation [7], angle QIM [8] and perceptually scaled QIM [9] restore scaling robustness by making the quantizer gain-adaptive. These schemes still write onto a (scaled) lattice; DP-QIM instead obtains scaling invariance as a by-product of matching the natural marginal, and our contribution is the joint behaviour of undetectability and robustness rather than scaling robustness per se. We do not compare against these gain-invariant variants (Sec. VII).

Learned watermarks. HiDDeN [10], StegaStamp [11], Stable Signature [12] and InvisMark [13] train encoder-

[1] https://github.com/sairam782/Invisible-Watermarking

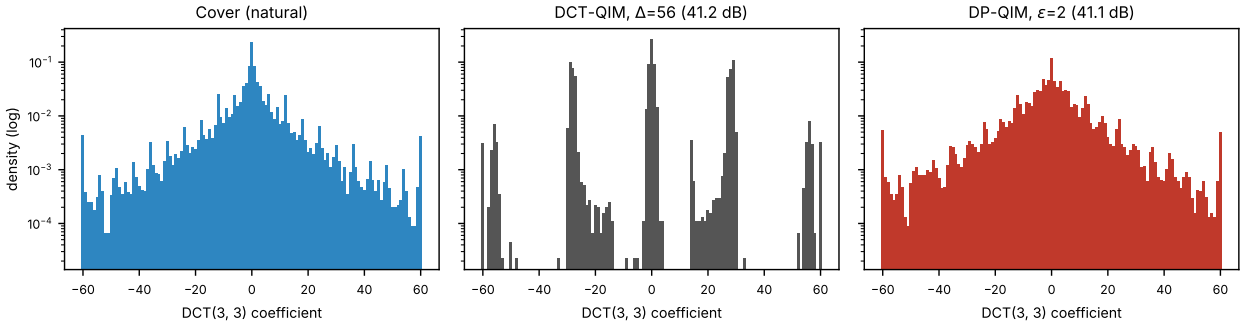


Fig. 1. Histogram (log scale) of DCT coefficient $(3, 3)$ over every block of the 44 real crops when every block carries one payload bit. Classical QIM (middle) collapses the coefficients onto lattice points; DP-QIM (right) is visually indistinguishable from the cover (left). Both watermarks are at matched PSNR (≈41 dB at this payload).

decoder networks against differentiable distortion layers. They are robust to common processing but remain vulnerable to regeneration [3] and to UnMarker [2]. Analogous statistical-signature concerns arise for language-model watermarks [14].

Steganalysis and distribution preservation. Steganography seeks to hide the *existence* of a message from statistical detectors [15], [16], and distribution-preserving embedding is a classical tool there. We borrow that goal for robust watermarking, where the adversary's aim is removal rather than detection, and measure undetectability with steganalysis-style detectors.

III. Distribution-Preserving QIM

A. Setting

An RGB image is converted to YCbCr (BT.601) and the luminance is split into 8×8 blocks with orthonormal 2-D DCT. A payload of P bits is carried by one coefficient position (u, v) in P blocks chosen by a keyed permutation over interior blocks (blocks touching reflect-padding are excluded because padding is re-derived from modified pixels at decode time). Both methods below use the same position and the same block selection.

DCT-QIM baseline. Dither-modulation QIM [5] with step Δ snaps the carrier c to $L_b = \{k\Delta + b\Delta/2\}$ and decodes by the nearer of L_0, L_1.

B. Embedding rule

Let F be the empirical CDF of coefficient (u, v) over a training set of natural images. For $M \geq 1$, quantile boundaries $F^{-1}(i/2M)$, $i = 0, \ldots, 2M$, define cells $\mathcal{C}_0, \ldots, \mathcal{C}_{2M-1}$ of equal natural mass with labels $\ell_i = i \bmod 2$. For a safety margin $\epsilon > 0$, the *safe pool* $\mathcal{P}_i$ holds training coefficients that lie in $\mathcal{C}_i$ at least ϵ from its finite boundaries. To embed bit b in a carrier c with cell index k:

1) if $\ell_k = b$, leave c unchanged;
2) otherwise, draw one keyed sample from each adjacent pool $\mathcal{P}_{k\pm1}$ and replace c with the one closer to c.

The decoder returns the label of the cell containing the received coefficient. All experiments use $M = 1$; larger M produced narrow cells near the mode that did not survive 8-bit rounding.

C. Properties

Proposition 1 (marginal)**.** *Let $M = 1$, payload bits be i.i.d. uniform, and F_ϵ be the natural distribution conditioned on lying at least ϵ from the cell boundary. Then the watermarked carrier distribution is $P_W = \frac{1}{2}F + \frac{1}{2}F_\epsilon$, and $\mathrm{KS}(P_W, F) = \frac{1}{2}\mathrm{KS}(F_\epsilon, F) \to 0$ as $\epsilon \to 0$.*

Sketch. Each cell carries mass $\frac{1}{2}$ and each bit matches the current label with probability $\frac{1}{2}$; matched carriers are untouched (distributed as F) and the rest are drawn from the target cell's safe pool, which over both cells is F_ϵ. KS is linear in a mixture weight against a common component. □

By contrast, QIM's carrier distribution is supported on a lattice, so its KS distance to F is bounded away from zero for any Δ.

Proposition 2 (scaling)**.** *Let an attack map the carrier $c \mapsto s\,c$ for some $s > 0$. (i) If the M=1 cell boundary is 0, the DP-QIM decision $\mathbb{1}\{c > 0\}$ is unchanged for every $s > 0$. (ii) A QIM carrier embedded at lattice value c decodes correctly after scaling iff $|1 - s|\,|c| < \Delta/4$; for $s \leq \frac{1}{2}$ every carrier with $|c| \geq \Delta/2$ lands on or beyond the decision boundary.*

Sketch. (i) $\mathrm{sign}(sc) = \mathrm{sign}(c)$. (ii) $L_0 \cup L_1$ has spacing $\Delta/2$, so decoding is correct iff the carrier moves by less than $\Delta/4$; the move is $|1 - s||c|$. □

The natural median of coefficient $(3, 3)$ over our data is exactly 0, so (i) applies. Linear low-pass operations act approximately as per-coefficient scaling: a Gaussian blur of standard deviation σ attenuates frequency f by $e^{-2\pi^2\sigma^2 f^2}$, which is 0.50 at the $(3, 3)$ basis frequency (f=3/16 cycles/pixel, σ=1) and 0.74 at $(2, 2)$. Prop. 2 predicts that DP-QIM should gain most where attacks scale strongly, which is what we observe (Sec. V-D).

IV. Evaluation Protocol

Data. We use the real photographs distributed with scikit-image and Matplotlib (outdoor and indoor scenes, a portrait,

natural textures, and astronomical and microscopy images; synthetic and text images excluded): 15 sources, cut into 44 non-overlapping 256×256 crops (at most four per source).

Splits. Five folds are grouped by *source*, so no test crop shares a photograph with any training crop. DP-QIM's CDF and pools, and the attacker's natural reference, are fitted on training sources only. Every crop is tested exactly once.

Matched fidelity. Both methods embed $P{=}64$ bits at the same position (main results at $(3,3)$). We sweep DP-QIM's margin $\epsilon \in \{0.5, 1, 2, 3, 4, 6\}$ and QIM's step $\Delta \in \{6, \ldots, 128\}$ (15 values spanning 46 to 72 dB), and compare DP-QIM against the QIM curve linearly interpolated to DP-QIM's mean PSNR. We never extrapolate beyond the sweep. Our main operating point is $\epsilon{=}2$ (53.9 dB), the smallest margin that decodes without attack (accuracy 0.999; $\epsilon{=}0.5$ and 1 give 0.949 and 0.976).

Attacks. JPEG ($Q{=}90, 70$), additive Gaussian noise ($\sigma{=}0.03$ of full range), Gaussian blur ($\sigma{=}1$), $0.5\times$ bicubic down-and-up-scaling, and contrast scaling about the image mean by 0.5 and 0.8 (pure valumetric scaling of all AC coefficients).

UnMarker-style removal. We reimplement the attack's mechanism from its description [2]: for each of the 63 AC positions the attacker compares the received image's coefficient histogram with a natural reference built from *training-source* images, flags bins whose density exceeds the reference by a ratio τ, and adds Gaussian jitter of j times the reference inter-quartile range to coefficients in flagged bins. The attacker never sees the embedder's pools or key. Attack strength is identical on both methods (attacked-image PSNR 47.0 dB for $\tau{=}2, j{=}0.5$ and 35.7 dB for $\tau{=}1.5, j{=}1$).

Detectors. (a) A learned detector: logistic regression on the 1-unit-bin histogram of the carrier position over *all* blocks of an image. Following Kerckhoffs' principle it knows the method and position, but not the key, so it cannot isolate carrier blocks. (b) The unsupervised UnMarker statistic: the maximum over-density of the image's histogram relative to a reference built from other sources. Both are scored by ROC-AUC under source-grouped five-fold evaluation, at $P \in \{64, 256, 1024\}$ (1024 = every block), with QIM's Δ chosen to match DP-QIM's PSNR at each payload.

Statistics. All 95% confidence intervals resample whole source photographs (2,000 bootstrap replicates). Payload bits are seeded by CRC32 of the crop name, so every run is reproducible.

V. Results

A. Detectability

Table I reports detectability. Against QIM, the learned detector is near-perfect from 256 bits on, and already reaches 0.90 at 64 bits even though only 6% of blocks carry payload. Against DP-QIM it is at or near chance at 64 and 256 bits, and its confidence intervals do not overlap QIM's. The UnMarker peak statistic, which is the cue a statistical removal attack relies on, is at chance for DP-QIM at 64 and 256 bits and rises only to 0.59 at full density, against 0.72 to 0.99 for QIM. At full density the learned detector does gain some purchase on DP-QIM (0.77), consistent with Prop. 1: the safe-interior pool removes mass near the cell boundary, a deficit that becomes measurable when every block is rewritten.

TABLE I
Detector ROC-AUC with 95% source-bootstrap CI (0.5 = chance). DP-QIM $\epsilon{=}2$ against QIM at matched PSNR, position $(3,3)$.

	Learned detector		Peak statistic	
Bits (PSNR)	DCT-QIM	DP-QIM	DCT-QIM	DP-QIM
64 (53.9)	0.90 [0.78, 1.00]	**0.52 [0.49, 0.57]**	0.72 [0.58, 0.88]	**0.49 [0.47, 0.49]**
256 (47.2)	1.00 [1.00, 1.00]	**0.61 [0.54, 0.71]**	0.86 [0.71, 1.00]	**0.47 [0.41, 0.50]**
1024 (41.1)	1.00 [0.99, 1.00]	**0.77 [0.61, 0.98]**	0.99 [0.98, 1.00]	**0.59 [0.46, 0.73]**

TABLE II
Bit accuracy of DP-QIM ($\epsilon{=}2$, 53.9 dB, position $(3,3)$, 64 bits) and its difference from PSNR-matched QIM. Bold: 95% CI excludes zero.

Attack	DP-QIM	Δ vs. DCT-QIM (pp)	95% CI
No attack	0.999	-0.1	[-0.3, +0.0]
JPEG $Q{=}90$	0.839	**-15.2**	[-19.2, -11.8]
JPEG $Q{=}70$	0.710	**-28.4**	[-31.4, -25.3]
Gaussian noise $\sigma{=}0.03$	0.829	**-13.6**	[-16.0, -11.6]
Gaussian blur $r{=}1$	0.837	**+34.6**	[+30.3, +39.6]
Downscale $0.5\times$	0.831	**+34.0**	[+30.5, +37.4]
Contrast $\times0.5$	0.967	**+19.6**	[+13.5, +30.0]
Contrast $\times0.8$	0.975	-0.3	[-2.5, +2.0]
UnMarker-style ($\tau{=}2$, $j{=}0.5$)	0.972	**-2.8**	[-5.2, -1.0]
UnMarker-style ($\tau{=}1.5$, $j{=}1$)	0.912	**-6.9**	[-9.6, -3.9]

B. Robustness at matched fidelity

Table II and Fig. 2 show a clear split. Under attacks that scale the carrier (Gaussian blur, downscaling and strong contrast reduction), DP-QIM retains 0.83 to 0.97 accuracy while matched QIM is near chance, a gain of $+20$ to $+35$ pp. Mild contrast scaling ($\times0.8$) moves QIM carriers by less than $\Delta/4$ and leaves both methods intact, as Prop. 2(ii) predicts. Under JPEG and additive noise the ordering reverses (-14 to -28 pp): at matched PSNR, QIM concentrates its distortion into a large step ($\Delta \approx 52$ here), which buys margin against perturbations that do not scale, whereas DP-QIM's carriers are ordinary natural values whose distance to the cell boundary is often small.

C. Removal attack

The UnMarker-style attack reduces DP-QIM to 0.97 ($\tau{=}2, j{=}0.5$) and 0.91 ($\tau{=}1.5, j{=}1$) bit accuracy, 3 and 7 pp below matched QIM. Because DP-QIM shows no histogram peaks, the attack's flagged bins mostly reflect image content, and its jitter then acts as additive noise, the regime in which QIM's larger margin wins. Undetectability thus does not by itself translate into removal resistance. What it does deny the attacker is the ability to tell watermarked from unwatermarked images (Table I), so a removal attack cannot be applied selectively and must degrade all content it touches.

D. Effect of position

Table III repeats the comparison at $(2,2)$ and $(4,3)$. The blur and pure-scaling gains hold at every position ($+20$ to

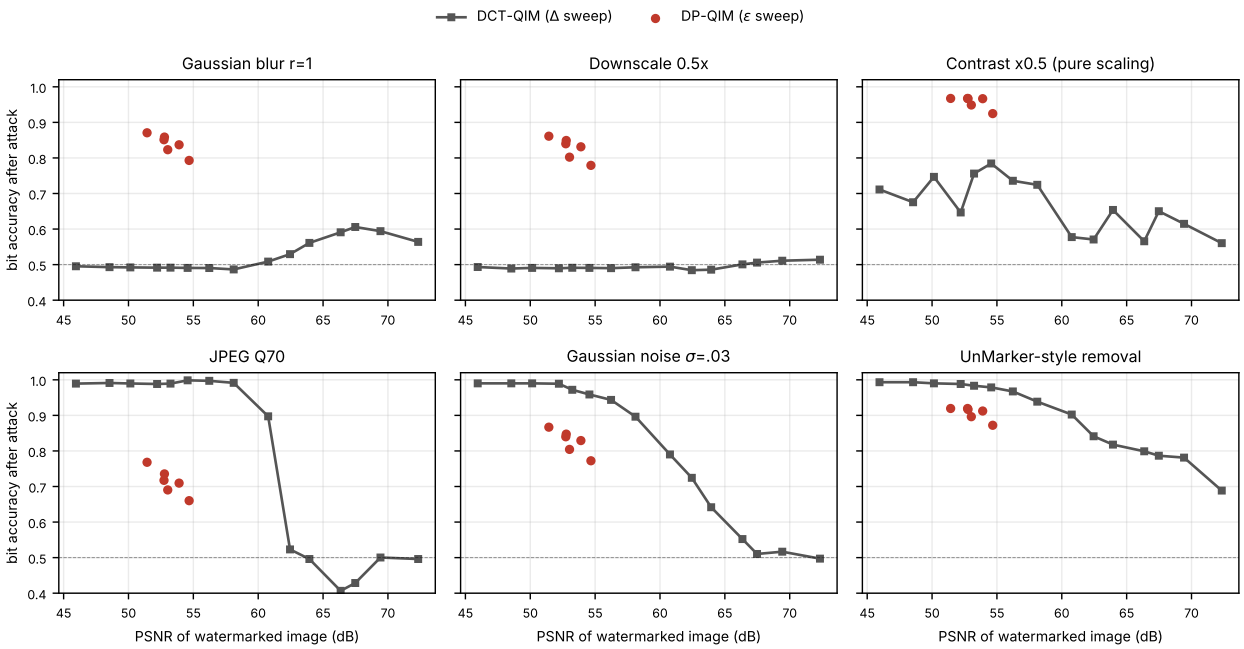


Fig. 2. Bit accuracy after attack versus PSNR of the watermarked image (44 real crops, position $(3, 3)$, 64 bits). QIM is a curve over its step Δ; DP-QIM points are its margin sweep. DP-QIM dominates under scaling-type attacks (top row) and is dominated under requantization and additive noise (bottom row).

TABLE III
GAIN (PP) OF DP-QIM (ϵ=2) OVER PSNR-MATCHED QIM AT THREE DCT POSITIONS. BOLD: 95% CI EXCLUDES ZERO.

Position	PSNR	Blur	Down. 0.5×	Contr. ×0.5	JPEG 70	Noise	UnMarker
$(2, 2)$	49.3	**+31**	**-6**	**+32**	**-17**	**-14**	**-6**
$(3, 3)$	53.9	**+35**	**+34**	**+20**	**-28**	**-14**	**-7**
$(4, 3)$	55.6	**+29**	**+7**	**+27**	-12	**-14**	**-8**

+35 pp), and the noise and requantization costs persist. The downscaling gain depends on position: it is largest at $(3, 3)$, small at $(4, 3)$, where downscaling attenuates the carrier so strongly that DP-QIM's sign is itself lost, and reverses at $(2, 2)$, where attenuation is mild enough for QIM to survive. This position dependence is why the baseline must be evaluated at the same position as the proposed method; in an earlier version of our own pipeline the baseline sat at a higher frequency than DP-QIM, which inflated the apparent gains.

VI. DISCUSSION

When to use DP-QIM. DP-QIM is preferable where an adversary should not be able to tell which images are marked, or where marked images are likely to be resized, filtered or tone-adjusted; QIM remains preferable where JPEG recompression and noise dominate and detectability is irrelevant. The two are complementary, which suggests hybrids that carry part of a payload with each; we leave this to future work.

Evaluation pitfalls. Rebuilding our evaluation exposed errors that we believe are easy to make: comparing methods at different DCT positions; building the attacker's "natural" reference from the embedder's own samples, which makes the defense look invisible by construction; computing per-image KS statistics on 64 samples, whose null distribution already has mean ≈ 0.11; fitting statistics on test images; and seeding payloads with Python's per-process salted `hash()`. Each changed our conclusions. The released protocol guards against all of them, and we recommend it for future comparisons of quantization watermarks.

VII. LIMITATIONS

- **Data.** 44 crops from 15 photographs, all distributed with open software; we could not run Kodak or DIV2K in our build environment. The loaders accept local folders, and these benchmarks are the immediate next step.
- **Baselines.** We compare only with lattice QIM. Gain-invariant QIM variants [7], [8], [9] would likely close the scaling gap while keeping a lattice fingerprint, and learned watermarks [10], [11] use different carriers; both comparisons are needed.
- **Attacks and detectors.** Our UnMarker-style attack follows the published description, not the authors' code, and we do not evaluate regeneration attacks. Our detectors are histogram-based; rich-model or CNN steganalyzers [16] may detect DP-QIM better.
- **Capacity and fidelity.** One bit per block at a single position; $M > 1$ was numerically fragile. Because a moved carrier is replaced by a random natural sample, DP-QIM's distortion is higher than necessary and its operating range at 64 bits is 51 to 55 dB, whereas QIM reaches above 70 dB. Sampling the replacement conditionally on the original magnitude could raise fidelity without breaking Prop. 1.

VIII. Conclusion

Constraining a QIM-style watermark to preserve the natural coefficient distribution makes it effectively invisible to histogram-based detection and, for one-bit cells split at a zero median, invariant to amplitude scaling. On real photographs at matched fidelity this yields large gains under blur, downscaling and contrast reduction, at the cost of margin against JPEG, additive noise and an UnMarker-style removal attack. Matching the natural distribution is therefore a real lever for watermark security, but not a free one, and it does not by itself defeat removal.

Reproducibility

All results are produced by the released code (https://github.com/sairam782/Invisible-Watermarking):

```
python -m experiments.eval_v2 curves --uv 3 3
python -m experiments.eval_v2 curves --uv 2 2
python -m experiments.eval_v2 curves --uv 4 3
python -m experiments.detect_v2
python -m experiments.tables_v2
python -m experiments.figures_v2
```

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